

Men o'quv markazi uchun to'liq CRM tizim yaratmoqchiman. Quyidagi BARCHA talablarni o'z ichiga olgan, matematik formulalarni qo'llab-quvvatlaydigan, lokatsiya xizmati orqali davomatni nazorat qiladigan va Telegram orqali avtomatik xabar yuboradigan to'liq veb-ilova yozib bering.

## ## 1. TEXNOLOGIYALAR

- Backend: Python (FastAPI) yoki Node.js (Express) — o'zing tanlagan, lekin Python FastAPI afzal
- Frontend: React.js + Bootstrap/Tailwind yoki oddiy HTML/CSS/JS
- Ma'lumotlar bazasi: PostgreSQL (asosiy) yoki SQLite (development)
- Formulalar uchun: LaTeX + MathJax yoki KaTeX
- Diagrammalar uchun: Chart.js yoki ECharts
- Geolokatsiya: Geolocation API + Haversine formulasi
- Telegram: python-telegram-bot (aiogram) yoki node-telegram-bot-api
- Xabar navbati: Celery + Redis (yoki node-cron)
- Eksport: openpyxl (Excel), pdfkit (PDF)
- Xavfsizlik: JWT authentication, bcrypt parol hashlash

## ## 2. ASOSIY MODULLAR

### ### 2.1. AUTHENTICATION VA ROLLAR

- Admin (to'liq huquq)
- O'qituvchi (test yaratish, natijalar, davomat)
- O'quvchi (faqat o'z natijalarini ko'rish, test topshirish, davomat belgilash)
- Ota-ona (faqat farzandining natijalarini ko'rish)
- Login/Register (email/telefon + parol)
- JWT token orqali autentifikatsiya
- Parolni tiklash (email orqali)

### ### 2.2. O'QUVCHILAR MODULI

- O'quvchi qo'shish, tahrirlash, o'chirish (faqat admin va o'qituvchi)
- O'quvchi ma'lumotlari:
  - ism, familiya, sharif
  - tug'ilgan sana
  - guruh (group\_id)

- telefon raqami
- ota-ona telefon raqami
- ota-ona Telegram ID (chat\_id)
- o'quvchi Telegram ID (chat\_id)
- qo'shilgan sana (enrolled\_date)
- kurs boshlanish va tugash sanasi (course\_start\_date, course\_end\_date)
- elektron pochta (email)
- manzil (uy manzili)
- rasm (avatar)
- O'quvchini qidirish (ism, telefon, guruh bo'yicha)
- O'quvchi statistikasi paneli:
  - Haftalik, oylik, kurs davomidagi ko'rsatkichlar
  - Progress (foizda o'sish)
  - O'rtacha ball
  - Eng yaxshi va eng yomon natijalar

### ### 2.3. GURUHLAR MODULI

- Guruh yaratish, tahrirlash, o'chirish
- Guruh ma'lumotlari:
  - nomi
  - tavsifi
  - o'quv markazi manzili (lokatsiya uchun)
  - Telegram guruhi linki
  - dars jadvali (kunlar va vaqtlar)
  - o'qituvchi (teacher\_id)
  - boshlanish va tugash sanasi
- Guruhga o'quvchilarni birlashtirish va ajratish
- Guruhdagi o'quvchilarning umumiy statistikasi
- Guruh bo'yicha hisobotlar

### ### 2.4. TESTLAR MODULI (MATEMATIK FORMULALAR BILAN)

- **\*\*Test yaratish\*\*** (o'qituvchi uchun):

- Test nomi va tavsifi
- Guruh tanlash (yoki barcha guruhlar uchun)
- Test turi: haftalik, oylik, yakuniy, amaliyot
- Davomiyligi (daqiqalarda)
- Savollar soni
- Har bir savol:
  - Savol matni (matn + LaTeX formulalar)
  - 4 ta variant (A, B, C, D) — har biri LaTeX formatida
  - To'g'ri javob (A/B/C/D)
  - Savol uchun ball (1-10)
- Formula muharriri:
  - Tugmalar:  $\frac{a}{b}$ ,  $\sqrt{x}$ ,  $x^2$ ,  $x_n$ ,  $\int$ ,  $\sum$ ,  $\prod$ ,  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\pi$ ,  $\infty$ ,  $\leq$ ,  $\geq$ ,  $\neq$ ,  $\cdot$ ,  $\times$ ,  $\div$ ,  $\mu$ ,  $\nu$
  - Tugma bosilganda LaTeX kodi inputga qo'shiladi
  - Real vaqtda MathJax preview (quyida ko'rsatiladi)
  - Variantlar uchun ham xuddi shunday
- Testni PDF yoki qog'ozga chiqarish (chop etish) tugmasi
- **Test topshirish** (o'quvchi uchun):
  - O'quvchi faqat o'z guruhiga mos testlarni ko'radi
  - Har bir savolda LaTeX formulalar MathJax bilan chiroyli ko'rsatiladi
  - Javoblar variantlar orqali tanlanadi
  - Taymer (qolgan vaqt ko'rsatiladi)
  - Vaqt tugaganda avtomatik yakunlanadi
  - Testni saqlash va keyin davom ettirish (draft holati)
- **Avtomatik tekshirish**:
  - Test tugagach, natija avtomatik hisoblanadi
  - Har bir savol bo'yicha batafsil natija: ☒ to'g'ri / ☒ xato
  - To'g'ri javoblar soni / umumiy savollar soni (foizda)
  - Umumiy ball
  - Test natijasi darajasi: "A'lo" (90%+), "Yaxshi" (75-89%), "O'rtacha" (50-74%), "Yomon" (50% dan past)
  - Xatolar bo'yicha tavsiyalar (qaysi mavzularni takrorlash kerak)
- **Test natijalarini boshqarish**:
  - Barcha o'quvchilarning natijalarini ko'rish

- Natijalarni Excel, PDF, CSV ga eksport qilish
- O'quvchilarni natijalar bo'yicha saralash

### ### 2.5. SAVOL-JAVOB MODULI (Dars oldi)

- O'qituvchi savol va to'g'ri javobni kiritadi (LaTeX formatida)
- Savol turi: test (variantli) yoki ochiq (matnli)
- O'quvchilar o'z javoblarini kiritadilar (variant tanlash yoki matn yozish)
- Tizim avtomatik tekshiradi (variantli) yoki o'qituvchi tekshiradi (ochiq)
- Har bir o'quvchi uchun to'g'ri javoblar soni / umumiy savollar soni (foizda)
- Kunlik, haftalik, oylik statistika
- Savol-javob sessiyasini boshlash va yakunlash

### ### 2.6. FAOLLIK MODULI

- O'qituvchi har bir o'quvchiga darsdagi faolligi uchun ball qo'yadi (0-10)
- Faollik mezonlari: savol berish, javob berish, doskada ishlash, guruhda ishlash
- O'rtacha faollik foizi hisoblanadi
- Faollik grafigi (kun/oy bo'yicha)
- Eng faol o'quvchilar reytingi

### ### 2.7. DAVOMAT MODULI (LOKATSIYA XIZMATI BILAN)

- **\*\*O'quv markazi manzili\*\***:
  - CRM tizimida o'quv markazining aniq manzili saqlanadi
  - GPS koordinatalari (latitude, longitude)
  - Ruxsat etilgan radius (metrda, default 100 metr)
  - Virtual chegara (Geo-fencing)
- **\*\*O'quvchi tomonidan davomat belgilash\*\***:
  - O'quvchi mobil ilova yoki veb-sayt orqali "Davomatni belgilash" tugmasini bosadi
  - Tizim foydalanuvchining joriy lokatsiyasini so'raydi (Geolocation API)
  - Tizim Haversine formulasi yordamida masofani hisoblaydi:
  - Agar masofa  $\leq$  ruxsat etilgan radius bo'lsa ☒ "Keldi" holati
  - Agar masofa  $>$  ruxsat etilgan radius bo'lsa ☒ "Kelmadi" holati (yoki "Kechikdi")
  - Agar o'quvchi dars boshlanish vaqtidan kechikib kelsa, "Kechikdi" va daqiqalar soni hisoblanadi

- Soxta lokatsiyani oldini olish uchun qurilma ID si (device\_id) saqlanadi
- **\*\*Yo'qlama jadvali\*\***:
- Har bir guruh uchun kunlik yo'qlama jadvali
- Ustunlar: O'quvchi, Holat (Keldi/Kechikdi/Kelmadi), Kechikish (daqqa), Lokatsiya (manzil), Tasdiqlash vaqti, Qurilma ID
- Lokatsiya xaritada ko'rsatiladi (Google Maps yoki OpenStreetMap)
- Davomat foizi: (kelgan kunlar / umumiy kunlar) \* 100
- **\*\*O'qituvchi tomonidan qo'lda tahrirlash\*\***:
- O'qituvchi agar texnik muammo bo'lsa, qo'lda davomat holatini o'zgartirishi mumkin
- O'zgartirish tarixi (audit log) saqlanadi

### ### 2.8. NATIJALAR VA DIAGRAMMALAR MODULI

- **\*\*Har bir o'quvchi uchun individual diagrammalar\*\***:

- 3 ta doiraviy diagramma (pie chart):

1. Savol-javob natijasi (to'g'ri/xato foizlari)
2. Test natijasi (to'g'ri/xato foizlari)
3. Faollik (ball / maksimal ball foizi)

- Line chart (chiziqli grafik):

4. Haftalik progress (7 kunlik)
5. Oylik progress (30 kunlik)
6. Kurs davomidagi progress

- Bar chart (ustunli grafik):

7. Haftalik test natijalari
8. Oylik test natijalari
9. Mavzular bo'yicha natijalar

- **\*\*Diagrammalar xususiyatlari\*\***:

- Rasm sifatida eksport (PNG/SVG)
- Interaktiv (hover ma'lumotlar)
- Ranglar moslashtirilgan
- Chart.js yoki ECharts bilan chizilgan

- **\*\*Hisobot sahifasi\*\***:

- Barcha diagrammalar bir sahifada
- "Hisobotni yuklab olish" tugmasi (PDF)

- Vaqt oralig'ini tanlash (kun, hafta, oy, kurs)

#### ### 2.9. SHAXSIY HISOBOTLAR

- **\*\*Haftalik hisobot\*\***:

- Haftadagi barcha test natijalari
- Savol-javob natijalari
- Faollik ballari
- Davomat (kelgan/kechikkan/kelmagan kunlar)
- O'rtacha ball
- Haftalik progress (oldingi haftaga nisbatan o'sish/kamayish)
- Hafta eng yaxshi va eng yomon natijasi
- "Hafta o'quvchisi" (eng yaxshi natija ko'rsatgan o'quvchi)

- **\*\*Oylik hisobot\*\***:

- Oydagi barcha test natijalari
- O'rtacha savol-javob natijasi
- O'rtacha faollik
- Davomat foizi (oy davomida)
- Oylik progress (oldingi oyga nisbatan)
- Oydagi eng yaxshi va eng yomon natijalar
- "Oy o'quvchisi"

- **\*\*Kurs davomidagi hisobot\*\***:

- Kurs boshidan beri barcha natijalar
- Umumiy o'rtacha ball
- Umumiy progress (boshidan oxirigacha grafik)
- Kurs yakunidagi umumiy baho
- Sertifikat yoki tavsiyanoma uchun ma'lumotlar
- "Kurs eng yaxshi o'quvchisi"

- **\*\*Shaxsiy kabinet\*\*** (o'quvchi va ota-ona uchun):

- O'quvchi o'z natijalarini ko'radi
- O'quvchi o'z progressini kuzatadi
- O'quvchi kelgusi testlar va darslar haqida ma'lumot oladi
- O'quvchi o'qituvchiga savol berishi mumkin

### ### 2.10. AVTOMATIK TELEGRAM XABAR

- **\*\*Testdan so'ng avtomatik xabar\*\*** (darhol yuboriladi):

- Xabar tarkibi:

👋 Salom [O'quvchi ismi]!

📄 Siz "[Test nomi]" testini topshirdingiz.

📊 Natijangiz: [to'g'ri javoblar]/[umumiy savollar] = [foiz]%

📋 Har bir savol bo'yicha natijangiz:

1 [savol qisqacha] — ☒ To'g'ri / ☒ Xato (To'g'ri javob: [A/B/C/D])

2 [savol qisqacha] — ☒ To'g'ri / ☒ Xato (To'g'ri javob: [A/B/C/D])

...

★ Umumiy ball: [ball]

🏆 Daraja: [A'lo/Yaxshi/O'rtacha/Yomon]

📖 Tavsiya: [zaif joylar bo'yicha maslahat]

👨👩 Ota-onangizga ham xabar yuborildi.

- Xabar kimga yuboriladi:

- O'quvchining o'ziga (agar Telegram ID bo'lsa)

- Ota-onasiga (agar Telegram ID bo'lsa)

- Ikkalasiga ham (default)

- **\*\*Haftalik avtomatik hisobot\*\*** (har yakshanba soat 18:00):

- Xabar tarkibi:

👨👩 Hurmatli [Ota-ona],  
farzandingiz [O'quvchi ismi]ning 📅 haftalik hisoboti:

📊 Testlar: [o'rtacha foiz]% (oxirgi test: [foiz]%)

💬 Savol-javob: [o'rtacha foiz]%

★ Faollik: [o'rtacha foiz]%

📅 Davomat: [foiz]% ([kelgan]/[umumiy] kun)

📈 O'tgan haftaga nisbatan: [o'sish/kamayish]%

🏆 Eng yaxshi natija: [sana] - [foiz]%

🗺️ Diagrammalar (ilovada)

Yaxshi hafta o'tdi! Davom etamiz 🔄

- 3 ta asosiy diagramma (pie chart) rasm sifatida ilova qilinadi

- **\*\*Oylik avtomatik hisobot\*\*** (har oy 1-kuni soat 10:00):
- Batafsil oylik hisobot (10+ ko'rsatkich)
- Oylik diagrammalar
- "Oy o'quvchisi" e'lon qilinadi
- **\*\*Kurs yakunida hisobot\*\***:
- Umumiy sertifikat
- Barcha ko'rsatkichlar
- O'qituvchi tavsiyasi
- **\*\*Telegram bot komandalari\*\***:
- `/start`` — ro'yxatdan o'tish va botga ulanish
- `/myresults`` — o'z natijalarini ko'rish (o'quvchi uchun)
- `/childresults`` — farzandining natijalarini ko'rish (ota-ona uchun)
- `/weekly`` — haftalik hisobotni olish
- `/monthly`` — oylik hisobotni olish
- `/course`` — kurs davomidagi hisobot
- `/attendance`` — davomatni ko'rish
- `/help`` — yordam
- **\*\*Xabarlar tarixi\*\***:
- Barcha yuborilgan xabarlar ma'lumotlar bazasida saqlanadi
- Kimga, qachon, qanday xabar yuborilgan

### ### 2.11. LOKATSIYA XIZMATI (Geo-fencing)

- **\*\*O'quv markazi manzilini saqlash\*\***:

```
```sql
```

```
CREATE TABLE center_location (
    id SERIAL PRIMARY KEY,
    address TEXT NOT NULL,
    latitude DECIMAL(10, 8) NOT NULL,
    longitude DECIMAL(11, 8) NOT NULL,
    allowed_radius_meters INTEGER DEFAULT 100
);
```

**Davomatga lokatsiya qo'shish:**

```
ALTER TABLE attendance
```

```
ADD COLUMN check_in_latitude DECIMAL(10, 8),
```

```
ADD COLUMN check_in_longitude DECIMAL(11, 8),
```

```
ADD COLUMN location_verified BOOLEAN DEFAULT FALSE,
```

```
ADD COLUMN device_id VARCHAR(255),
```

```
ADD COLUMN check_in_method VARCHAR(20) DEFAULT 'gps' CHECK (check_in_method IN ('gps',  
'manual', 'qr'));
```

**Lokatsiyani tekshirish funksiyasi** (Python):

```
from math import radians, sin, cos, sqrt, atan2
```

```
def haversine(lat1, lon1, lat2, lon2):
```

```
    R = 6371 # Yer radiusi (km)
```

```
    lat1, lon1, lat2, lon2 = map(radians, [lat1, lon1, lat2, lon2])
```

```
    dlat = lat2 - lat1
```

```
    dlon = lon2 - lon1
```

```
    a = sin(dlat/2)**2 + cos(lat1) * cos(lat2) * sin(dlon/2)**2
```

```
    c = 2 * atan2(sqrt(a), sqrt(1-a))
```

```
    return R * c * 1000 # metrda
```

- **QR-kod orqali davomat** (muqobil):
  - O'qituvchi har darsda yangi QR-kod yaratadi
  - O'quvchi QR-kodni skanerlaydi
  - Tizim lokatsiyani ham tekshiradi (2 faktorli autentifikatsiya)

## 2.12. HISOBOTLAR VA EKSPORT

- **Guruh bo'yicha hisobot:**
  - Barcha o'quvchilarning natijalari jadvali
  - O'rtacha ball, test, savol-javob, faollik, davomat
  - Reyting (1-o'rin, 2-o'rin, ...)
- **Eksport formatlari:**
  - Excel (.xlsx) — openpyxl yordamida
  - PDF (.pdf) — pdfkit yoki reportlab yordamida
  - CSV (.csv)
  - Diagrammalarni rasm (.png, .svg) sifatida
- **Avtomatik hisobot:**
  - Cron job / Celery scheduler orqali

- Har hafta yakunida, har oy yakunida, kurs yakunida
- Admin va o'qituvchilarga email orqali yuboriladi

### 3.MA'LUMOTLAR BAZASI TO'LIQ TUZILMASI

-- Guruhlar

```
CREATE TABLE groups (  
    id SERIAL PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    description TEXT,  
    teacher_id INTEGER REFERENCES users(id),  
    center_latitude DECIMAL(10, 8),  
    center_longitude DECIMAL(11, 8),  
    allowed_radius INTEGER DEFAULT 100,  
    telegram_group_link VARCHAR(255),  
    schedule JSONB, -- {monday: '09:00-11:00', wednesday: '14:00-16:00'}  
    start_date DATE,  
    end_date DATE,  
    is_active BOOLEAN DEFAULT TRUE,  
    created_at TIMESTAMP DEFAULT NOW(),  
    updated_at TIMESTAMP DEFAULT NOW()  
);
```

-- O'quvchilar

```
CREATE TABLE students (  
    id SERIAL PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    middle_name VARCHAR(50),  
    birth_date DATE,  
    group_id INTEGER REFERENCES groups(id) ON DELETE SET NULL,  
    phone VARCHAR(20) UNIQUE,  
    email VARCHAR(100) UNIQUE,
```

```
parent_phone VARCHAR(20),
parent_telegram_id VARCHAR(50),
student_telegram_id VARCHAR(50),
address TEXT,
avatar_url VARCHAR(255),
enrolled_date DATE DEFAULT CURRENT_DATE,
course_start_date DATE,
course_end_date DATE,
is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);
```

-- Foydalanuvchilar (admin, o'qituvchi, o'quvchi, ota-ona)

```
CREATE TABLE users (
    id SERIAL PRIMARY KEY,
    username VARCHAR(50) UNIQUE NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(20) CHECK (role IN ('admin', 'teacher', 'student', 'parent')),
    student_id INTEGER REFERENCES students(id) ON DELETE CASCADE,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT NOW(),
    updated_at TIMESTAMP DEFAULT NOW()
);
```

-- Testlar

```
CREATE TABLE tests (
    id SERIAL PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    group_id INTEGER REFERENCES groups(id) ON DELETE CASCADE,
```

```
teacher_id INTEGER REFERENCES users(id),
test_type VARCHAR(20) CHECK (test_type IN ('weekly', 'monthly', 'final', 'practice')),
duration_minutes INTEGER DEFAULT 30,
total_questions INTEGER DEFAULT 0,
passing_score INTEGER DEFAULT 50,
is_published BOOLEAN DEFAULT FALSE,
start_date TIMESTAMP,
end_date TIMESTAMP,
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);
```

-- Test savollari

```
CREATE TABLE test_questions (
    id SERIAL PRIMARY KEY,
    test_id INTEGER REFERENCES tests(id) ON DELETE CASCADE,
    question_text TEXT NOT NULL,
    option_a TEXT NOT NULL,
    option_b TEXT NOT NULL,
    option_c TEXT NOT NULL,
    option_d TEXT NOT NULL,
    correct_answer CHAR(1) CHECK (correct_answer IN ('A', 'B', 'C', 'D')),
    points INTEGER DEFAULT 1,
    question_order INTEGER,
    created_at TIMESTAMP DEFAULT NOW()
);
```

-- Test natijalari

```
CREATE TABLE test_results (
    id SERIAL PRIMARY KEY,
    student_id INTEGER REFERENCES students(id) ON DELETE CASCADE,
    test_id INTEGER REFERENCES tests(id) ON DELETE CASCADE,
```

```

correct_count INTEGER DEFAULT 0,
total_questions INTEGER DEFAULT 0,
percentage DECIMAL(5,2) DEFAULT 0,
total_points INTEGER DEFAULT 0,
answers JSONB, -- {'1': 'A', '2': 'B', '3': 'C'}
started_at TIMESTAMP,
submitted_at TIMESTAMP DEFAULT NOW(),
status VARCHAR(20) DEFAULT 'completed' CHECK (status IN ('draft', 'completed'))
);

```

-- Savol-javob sessiyalari

```

CREATE TABLE qa_sessions (
    id SERIAL PRIMARY KEY,
    group_id INTEGER REFERENCES groups(id) ON DELETE CASCADE,
    teacher_id INTEGER REFERENCES users(id),
    title VARCHAR(255),
    question TEXT NOT NULL,
    correct_answer TEXT,
    is_multiple_choice BOOLEAN DEFAULT TRUE,
    options JSONB, -- ['A: ...', 'B: ...', 'C: ...', 'D: ...']
    started_at TIMESTAMP DEFAULT NOW(),
    ended_at TIMESTAMP,
    is_active BOOLEAN DEFAULT TRUE
);

```

-- Savol-javob natijalari

```

CREATE TABLE qa_results (
    id SERIAL PRIMARY KEY,
    student_id INTEGER REFERENCES students(id) ON DELETE CASCADE,
    qa_session_id INTEGER REFERENCES qa_sessions(id) ON DELETE CASCADE,
    answer TEXT,
    is_correct BOOLEAN DEFAULT FALSE,

```

```
points INTEGER DEFAULT 0,  
answered_at TIMESTAMP DEFAULT NOW()  
);
```

-- Faollik

```
CREATE TABLE activity_logs (  
    id SERIAL PRIMARY KEY,  
    student_id INTEGER REFERENCES students(id) ON DELETE CASCADE,  
    teacher_id INTEGER REFERENCES users(id),  
    score INTEGER CHECK (score >= 0 AND score <= 10),  
    comment TEXT,  
    date DATE DEFAULT CURRENT_DATE,  
    created_at TIMESTAMP DEFAULT NOW()  
);
```

-- Davomat (lokatsiya bilan)

```
CREATE TABLE attendance (  
    id SERIAL PRIMARY KEY,  
    student_id INTEGER REFERENCES students(id) ON DELETE CASCADE,  
    group_id INTEGER REFERENCES groups(id),  
    date DATE DEFAULT CURRENT_DATE,  
    status VARCHAR(20) CHECK (status IN ('present', 'late', 'absent', 'excused')),  
    late_minutes INTEGER DEFAULT 0,  
    check_in_latitude DECIMAL(10, 8),  
    check_in_longitude DECIMAL(11, 8),  
    check_in_address TEXT,  
    location_verified BOOLEAN DEFAULT FALSE,  
    check_in_method VARCHAR(20) DEFAULT 'gps' CHECK (check_in_method IN ('gps', 'manual', 'qr',  
'qrcode')),  
    device_id VARCHAR(255),  
    distance_from_center DECIMAL(10, 2),  
    check_in_time TIMESTAMP DEFAULT NOW(),  
    created_at TIMESTAMP DEFAULT NOW(),
```

```

        updated_at TIMESTAMP DEFAULT NOW()
    );

-- Telegram xabarlar tarixi
CREATE TABLE telegram_messages (
    id SERIAL PRIMARY KEY,
    student_id INTEGER REFERENCES students(id) ON DELETE CASCADE,
    chat_id VARCHAR(50) NOT NULL,
    chat_type VARCHAR(20) CHECK (chat_type IN ('student', 'parent', 'group')),
    message_type VARCHAR(30) CHECK (message_type IN ('test_result', 'weekly_report',
'monthly_report', 'final_report', 'attendance_reminder', 'custom')),
    message_text TEXT,
    image_paths JSONB, -- ['/path/to/image1.png', '/path/to/image2.png']
    file_path VARCHAR(255),
    sent_at TIMESTAMP DEFAULT NOW(),
    is_sent BOOLEAN DEFAULT TRUE,
    error_message TEXT
);

-- Hisobotlar
CREATE TABLE reports (
    id SERIAL PRIMARY KEY,
    student_id INTEGER REFERENCES students(id) ON DELETE CASCADE,
    group_id INTEGER REFERENCES groups(id),
    report_type VARCHAR(20) CHECK (report_type IN ('weekly', 'monthly', 'course')),
    data JSONB,
    file_path VARCHAR(255),
    generated_at TIMESTAMP DEFAULT NOW()
);

-- Audit log
CREATE TABLE audit_logs (
    id SERIAL PRIMARY KEY,

```

```
user_id INTEGER REFERENCES users(id),
action VARCHAR(100),
table_name VARCHAR(50),
record_id INTEGER,
old_data JSONB,
new_data JSONB,
ip_address VARCHAR(45),
user_agent TEXT,
created_at TIMESTAMP DEFAULT NOW()
);
```

-- System settings

```
CREATE TABLE system_settings (
    id SERIAL PRIMARY KEY,
    key VARCHAR(100) UNIQUE NOT NULL,
    value JSONB,
    description TEXT,
    updated_at TIMESTAMP DEFAULT NOW()
);
```

-- Indeksler

```
CREATE INDEX idx_students_group_id ON students(group_id);
CREATE INDEX idx_students_phone ON students(phone);
CREATE INDEX idx_students_email ON students(email);
CREATE INDEX idx_tests_group_id ON tests(group_id);
CREATE INDEX idx_test_results_student_id ON test_results(student_id);
CREATE INDEX idx_test_results_test_id ON test_results(test_id);
CREATE INDEX idx_attendance_student_id ON attendance(student_id);
CREATE INDEX idx_attendance_date ON attendance(date);
CREATE INDEX idx_attendance_group_id ON attendance(group_id);
CREATE INDEX idx_qa_results_student_id ON qa_results(student_id);
CREATE INDEX idx_activity_logs_student_id ON activity_logs(student_id);
```

```
CREATE INDEX idx_telegram_messages_chat_id ON telegram_messages(chat_id);
```

## 4. API ENDPOINTLAR (TO'LIQ)

### AUTHENTICATION

POST /api/auth/login

POST /api/auth/register

POST /api/auth/logout

POST /api/auth/refresh-token

POST /api/auth/forgot-password

POST /api/auth/reset-password

GET /api/auth/me

### USERS

GET /api/users

POST /api/users

PUT /api/users/{id}

DELETE /api/users/{id}

GET /api/users/{id}/permissions

### STUDENTS

GET /api/students

POST /api/students

GET /api/students/{id}

PUT /api/students/{id}

DELETE /api/students/{id}

GET /api/students/{id}/statistics

GET /api/students/{id}/weekly-report

GET /api/students/{id}/monthly-report

GET /api/students/{id}/course-report

GET /api/students/search?q=keyword

POST /api/students/{id}/assign-telegram

GET /api/students/{id}/telegram-status

## GROUPS

GET /api/groups

POST /api/groups

GET /api/groups/{id}

PUT /api/groups/{id}

DELETE /api/groups/{id}

GET /api/groups/{id}/students

POST /api/groups/{id}/students

DELETE /api/groups/{id}/students/{student\_id}

GET /api/groups/{id}/statistics

GET /api/groups/{id}/attendance

GET /api/groups/{id}/tests

## TESTS

GET /api/tests

POST /api/tests

GET /api/tests/{id}

PUT /api/tests/{id}

DELETE /api/tests/{id}

POST /api/tests/{id}/publish

POST /api/tests/{id}/unpublish

GET /api/tests/{id}/questions

POST /api/tests/{id}/questions

PUT /api/tests/questions/{question\_id}

DELETE /api/tests/questions/{question\_id}

POST /api/tests/{id}/submit

GET /api/tests/{id}/results

GET /api/tests/{id}/results/{student\_id}

GET /api/tests/{id}/export/excel

GET /api/tests/{id}/export/pdf

POST /api/tests/{id}/print

## QA (SAVOL-JAVOB)

POST /api/qa/session

GET /api/qa/session/{id}

PUT /api/qa/session/{id}

POST /api/qa/session/{id}/end

POST /api/qa/answer

GET /api/qa/results/{student\_id}

GET /api/qa/results/session/{session\_id}

GET /api/qa/results/group/{group\_id}

## ACTIVITY

POST /api/activity

GET /api/activity/{student\_id}

GET /api/activity/group/{group\_id}

GET /api/activity/statistics/{student\_id}

PUT /api/activity/{id}

## ATTENDANCE (LOKATSIYA BILAN)

POST /api/attendance/check-in

GET /api/attendance/check-in/status

GET /api/attendance/student/{student\_id}

GET /api/attendance/group/{group\_id}

GET /api/attendance/group/{group\_id}/today

GET /api/attendance/statistics/{student\_id}

PUT /api/attendance/{id}

POST /api/attendance/qr/scan

GET /api/attendance/qr/generate/{group\_id}

GET /api/attendance/export/excel

GET /api/attendance/export/pdf

## CHARTS & DIAGRAMMAS

GET /api/charts/student/{student\_id}  
GET /api/charts/student/{student\_id}/weekly  
GET /api/charts/student/{student\_id}/monthly  
GET /api/charts/student/{student\_id}/course  
GET /api/charts/group/{group\_id}  
GET /api/charts/group/{group\_id}/weekly  
GET /api/charts/group/{group\_id}/monthly

## REPORTS

GET /api/reports/student/{student\_id}/weekly  
GET /api/reports/student/{student\_id}/monthly  
GET /api/reports/student/{student\_id}/course  
GET /api/reports/group/{group\_id}/weekly  
GET /api/reports/group/{group\_id}/monthly  
GET /api/reports/group/{group\_id}/course  
GET /api/reports/student/{student\_id}/export/pdf  
GET /api/reports/student/{student\_id}/export/excel  
GET /api/reports/group/{group\_id}/export/pdf  
GET /api/reports/group/{group\_id}/export/excel

## TELEGRAM

POST /api/telegram/bot/webhook  
POST /api/telegram/send  
POST /api/telegram/send/test-result/{result\_id}  
POST /api/telegram/send/weekly-report/{student\_id}  
POST /api/telegram/send/monthly-report/{student\_id}  
POST /api/telegram/send/final-report/{student\_id}  
POST /api/telegram/send/bulk  
GET /api/telegram/messages/{student\_id}  
GET /api/telegram/messages/chat/{chat\_id}  
GET /api/telegram/status/{student\_id}

## SYSTEM

GET /api/system/settings

PUT /api/system/settings

GET /api/system/audit-logs

GET /api/system/health

POST /api/system/backup

GET /api/system/statistics

## 5. FRONTEND SAHIFALAR (TO'LIQ RO'YXAT)

1. **Dashboard** — umumiy statistika, so'nggi faoliyatlar, grafikalar
2. **O'quvchilar** — ro'yxat, qo'shish, tahrirlash, o'chirish, qidirish
3. **O'quvchi shaxsiy kabineti:**
  - Shaxsiy ma'lumotlar
  - Haftalik hisobot
  - Oylik hisobot
  - Kurs davomidagi hisobot
  - Diagrammalar (barcha turlari)
  - Progress grafiklari
  - Testlar tarixi
  - Davomat tarixi (xaritada)
4. **Guruhlar** — ro'yxat, qo'shish, tahrirlash, o'chirish
5. **Guruh sahifasi:**
  - Guruh ma'lumotlari
  - O'quvchilar ro'yxati
  - Kunlik yo'qlama (lokatsiya bilan)
  - Guruh statistikasi
  - Dars jadvali
6. **Testlar** — ro'yxat, yaratish, tahrirlash, o'chirish
7. **Test yaratish:**
  - LaTeX muharriri (tugmalar + preview)
  - Variantlar kiritish
  - To'g'ri javob tanlash
  - Chop etish tugmasi
8. **Test topshirish** (o'quvchi):
  - Test ro'yxati
  - Taymer
  - Savollarni navigatsiya qilish
  - Natija sahifasi
9. **Savol-javob** — yangi sessiya, natijalar, statistika

10. **Faollik** — ball qo'shish, statistika, reyting

11. **Davomat:**

- O'quvchi uchun: "Davomatni belgilash" tugmasi (lokatsiya tekshiruvi)
- O'qituvchi uchun: yo'qlama jadvali, qo'lda tahrirlash
- Xaritada o'quvchilarning joylashuvi
- QR-kod generatsiya va skanerlash

12. **Diagrammalar** — barcha o'quvchilar uchun individual diagrammalar

13. **Hisobotlar** — guruh va shaxsiy hisobotlar, eksport

14. **Telegram xabarlar** — xabarlar tarixi, avtomatik yuborish sozlamalari

15. **Sozlamalar** — tizim sozlamalari, o'quv markazi manzili, lokatsiya radiusi

16. **Audit log** — tizimdagi barcha o'zgarishlar tarixi

## 6. TELEGRAM BOT TO'LIQ FUNKSIYALARI

# Bot komandolari

```
@bot.message_handler(commands=['start'])
```

```
async def start(message):
```

```
    # Foydalanuvchini ro'yxatdan o'tkazish
```

```
    pass
```

```
@bot.message_handler(commands=['register'])
```

```
async def register(message):
```

```
    # O'quvchi yoki ota-onani tizimga ulash
```

```
    pass
```

```
@bot.message_handler(commands=['myresults'])
```

```
async def my_results(message):
```

```
    # O'quvchi o'z natijalarini ko'rish
```

```
    # 3 ta pie chart + so'nggi test natijalari
```

```
    pass
```

```
@bot.message_handler(commands=['childresults'])
```

```
async def child_results(message):
```

```
    # Ota-ona farzandining natijalarini ko'rish
```

```
    pass
```

```
@bot.message_handler(commands=['weekly'])  
  
async def weekly_report(message):  
    # Haftalik hisobot (diagrammalar bilan)  
  
    pass
```

```
@bot.message_handler(commands=['monthly'])  
  
async def monthly_report(message):  
    # Oylik hisobot (diagrammalar bilan)  
  
    pass
```

```
@bot.message_handler(commands=['course'])  
  
async def course_report(message):  
    # Kurs davomidagi hisobot  
  
    pass
```

```
@bot.message_handler(commands=['attendance'])  
  
async def attendance(message):  
    # Davomatni ko'rish  
  
    pass
```

```
@bot.message_handler(commands=['testresult'])  
  
async def test_result(message):  
    # So'nggi test natijasi  
  
    pass
```

```
@bot.message_handler(commands=['help'])  
  
async def help_command(message):  
    # Yordam  
  
    pass
```

```
@bot.message_handler(commands=['settings'])  
  
async def settings(message):
```

```
# Xabar sozlamalari (qaysi xabarlarni olish)
```

```
pass
```

```
@bot.callback_query_handler(func=lambda call: True)
```

```
async def callback_handler(call):
```

```
    # Tugmachalar bilan interaktiv menyu
```

```
    pass
```

```
# Avtomatik yuborish funksiyalari
```

```
async def send_test_result(student_id, test_result_id):
```

```
    # Testdan so'ng avtomatik xabar
```

```
    pass
```

```
async def send_weekly_reports():
```

```
    # Har yakshanba 18:00 da barcha o'quvchilarga haftalik hisobot
```

```
    pass
```

```
async def send_monthly_reports():
```

```
    # Har oy 1-kuni 10:00 da oylik hisobot
```

```
    pass
```

```
async def send_course_reports():
```

```
    # Kurs yakunida umumiy hisobot
```

```
    Pass
```

## 7. CRON/SCHEDULER VAZIFALARI

```
python
```

```
# Celery yoki node-cron bilan quyidagi vazifalar:
```

```
# 1. Har yakshanba 18:00 - haftalik hisobot yuborish
```

```
# 2. Har oy 1-kuni 10:00 - oylik hisobot yuborish
```

```
# 3. Har darsdan 15 daqiqa oldin - eslatma xabari
```

```
# 4. Har kuni 00:00 - kunlik statistika hisoblash
```

```
# 5. Har dars boshlanishida - avtomatik yo'qlama holatini yangilash
```

# 6. Har hafta dushanba 08:00 - haftalik jadval eslatmasi  
# 7. Har oy 1-kuni 00:00 - oylik arxiv yaratish  
# 8. Har hafta yakunida - zaxira nusxa (backup) olish

## 8. XAVFSIZLIK TALABLARI

- Barcha parollar bcrypt bilan hashlanadi
- JWT tokenlar (access + refresh)
- CORS sozlamalari (faqat ruxsat etilgan domenlar)
- Rate limiting (100 so'rov/daqiqa)
- SQL injection dan himoya (ORM yoki parametrli so'rovlar)
- XSS dan himoya (sanitizatsiya)
- CSRF tokenlar
- HTTPS majburiy
- Lokatsiya ma'lumotlarini shifrlash
- Qurilma ID orqali ruxsatsiz kirishni oldini olish
- Audit log (barcha o'zgarishlar saqlanadi)

## 9. ISHGA TUSHIRISH VA O'RNATISH

### **.env fayl namunasi:**

text

```
DATABASE_URL=postgresql://user:password@localhost/crm_db
SECRET_KEY=your-secret-key
JWT_SECRET=your-jwt-secret
TELEGRAM_BOT_TOKEN=your-bot-token
REDIS_URL=redis://localhost:6379
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your-email@gmail.com
SMTP_PASSWORD=your-password
CENTERS_LATITUDE=41.2995
CENTERS_LONGITUDE=69.2401
ALLOWED_RADIUS=100
GOOGLE_MAPS_API_KEY=your-google-maps-api-key
```

# Docker Compose:

yaml

version: '3.8'

services:

db:

image: postgres:15

environment:

POSTGRES\_DB: crm\_db

POSTGRES\_USER: user

POSTGRES\_PASSWORD: password

volumes:

- postgres\_data:/var/lib/postgresql/data

ports:

- "5432:5432"

redis:

image: redis:7

ports:

- "6379:6379"

backend:

build: ./backend

environment:

- DATABASE\_URL=postgresql://user:password@db/crm\_db

- REDIS\_URL=redis://redis:6379

depends\_on:

- db

- redis

ports:

- "8000:8000"

volumes:

- ./backend:/app

- /app/node\_modules

frontend:

build: ./frontend

ports:

- "3000:3000"

depends\_on:

- backend

volumes:

- ./frontend:/app

- /app/node\_modules

**celery:**

**build:** ./backend

**command:** celery -A app.celery worker --loglevel=info

**depends\_on:**

- redis
- db

**environment:**

- DATABASE\_URL=postgresql://user:password@db/crm\_db
- REDIS\_URL=redis://redis:6379

**celery-beat:**

**build:** ./backend

**command:** celery -A app.celery beat --loglevel=info

**depends\_on:**

- redis
- db

**volumes:**

postgres\_data:

## 10. QO'SHIMCHA TALABLAR

- **Mobile responsive** — barcha sahifalar mobil qurilmalarda yaxshi ko'rinishi
- **Dark mode** qo'llab-quvvatlashi
- **Offline mode** — internet bo'lmaganda ham davomat belgilash (keyin sinxronlash)
- **Push notifications** — veb-push orqali eslatmalar
- **Multi-language** — o'zbek va rus tillari qo'llab-quvvatlanishi
- **Zaxira nusxa** — har kuni avtomatik backup
- **Dars jadvali** — Google Calendar bilan integratsiya
- **To'lov moduli** — o'quv to'lovlarini kuzatish (ixtiyoriy)

---

Ilтимos, yuqoridagi BARCHA talablarni o'z ichiga olgan to'liq ishlaydigan dastur kodini yozib bering. Har bir modulni alohida faylda, tushunarli va izohlar bilan yozing.

MUHIM TALABLAR (albatta bo'lishi kerak):

1. O'quvchining shaxsiy ma'lumotlari va ko'rsatkichlari (haftalik, oylik, kurs davomida)
2. Har bir testdan so'ng o'quvchi va ota-onasiga Telegram orqali natija yuborish (avtomatik)
3. Matematik formulalar bilan ishlash (LaTeX + MathJax) — vizual muharrir bilan
4. Avtomatik hisobotlar (haftalik, oylik, kurs yakuni) — Telegram orqali
5. Lokatsiya xizmati orqali davomatni nazorat qilish (Geo-fencing + Haversine formulasi)
6. Kunlik guruh yo'qlamasida lokatsiya ma'lumotlari aks etishi
7. Diagrammalar (pie chart, line chart, bar chart) — har bir o'quvchi uchun individual
8. Barcha hisobotlarni Excel, PDF, CSV ga eksport qilish
9. Audit log — barcha o'zgarishlar tarixi
10. Docker orqali ishga tushirish